

Technical University
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Department of Civil and Mechanical Engineering

41525 - Finite Element Methods

Assignment 2

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Exercise 1 and code - Ludvig August Grønlien-Magnussen
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Exercise 4 and code - Martin Hoel

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1 Exercise 1 - Equation solving

1.1 Matrix generation and format comparison

We analysed a cantilever beam (length-to-height ratio 5:1) fixed at the left edge and loaded by a surface pressure $p = 1$ in plane stress with 4-node elements, using $E = 10000$, $\nu = 0.3$, $\rho = 1$ and $t = 1$. Eight structured meshes ($n_x \times n_y$) with 2048–5000 elements were generated in FlExtract, each element count in two orientations to show the effect of the column-wise node numbering (Table 1). Bandwidths were minimized with `bandfem`.

Table 1: Mesh data and compliance $C = \mathbf{p}^T\mathbf{d}$.

Mesh	Ne	Neqn	Bw	$\text{Bw}_{reord.}$	C
64×32	2048	4290	70	70	0.200042
60×50	3000	6222	106	166	0.199991
100×40	4000	8282	86	86	0.200389
100×50	5000	10302	106	106	0.200399
32×64	2048	4290	134	70	0.198504
50×60	3000	6222	126	166	0.199725
40×100	4000	8282	206	86	0.199247
50×100	5000	10302	206	106	0.199732

1.2 Storage requirement analysis

Full storage grows as n_{eqn}^2 and banded storage as $n_{eqn} \cdot bw$, giving large savings that increase with problem size. Since FlExtract numbers the nodes column-wise, the bandwidth is determined by the number of nodes per column. When n_y is clearly smaller than n_x, the original numbering already gives a near-minimal bandwidth, and `bandfem` leaves it unchanged, see Table 1

Table 2: Stored entries ($\times 10^6$) and saving $(1 - bw/n_{eqn}) \cdot 100\,\%$.

	Full	Bw		$\text{Bw}_{reord.}$	
Mesh	entries	entries	saving	entries	saving
64×32	18.40	0.300	98.37 %	0.300	98.37 %
60×50	38.71	0.660	98.30 %	1.033	97.33 %
100×40	68.59	0.712	98.96 %	0.712	98.96 %
100×50	106.13	1.092	98.97 %	1.092	98.97 %
32×64	18.40	0.575	96.88 %	0.300	98.37 %
50×60	38.71	0.784	97.97 %	1.033	97.33 %
40×100	68.59	1.706	97.51 %	0.712	98.96 %
50×100	106.13	2.122	98.00 %	1.092	98.97 %

1.3 CPU time benchmarking

Assembly and solution were timed separately using the CPU time, see Figure 1.2.1. Repeated runs of the same mesh gave slightly different times due to power settings and background processes, while the differences between meshes are determined by the bandwidth and problem size. For large meshes numbered along the long side, reordering reduced the bandwidth and the band solve time, e.g. from 0.64 s to 0.17 s for the 50×100 mesh, while the assembly was unaffected. The compliance was identical before and after reordering, confirming that only the ordering of the equations changes, not the solution. For both formats the factorization dominates the solve time, since it requires about $2n_{eqn}^3/3$ operations for full storage and $b_w^2 n_{eqn}/2$ for banded storage, while the forward and backward substitutions only require about $2n_{eqn}^2$ and $2b_w n_{eqn}$, respectively.

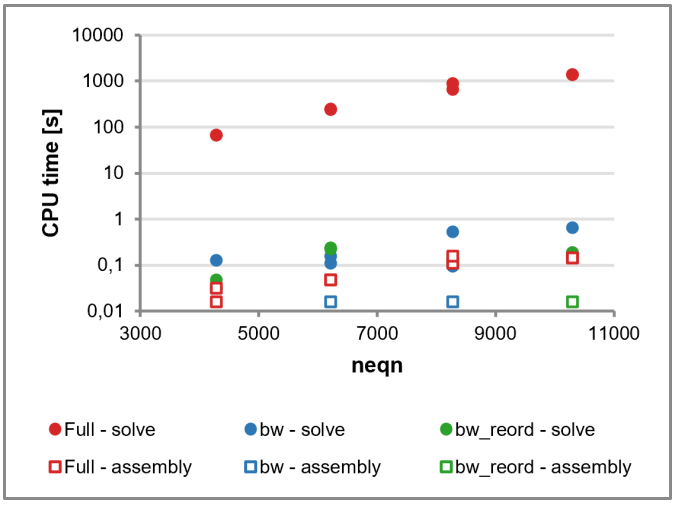


Figure 1.2.1: CPU time benchmarking

For full storage, the theory predicts the solve time to increase by $(\frac{10302}{4290})^3 \approx 14$ times from the smallest to largest mesh, while we measured about $\frac{1372}{66.6} \approx 21$ times. For banded storage, reordering reduced b_w from 206 to 106, which should reduce the solve time by $(206/106)^2 \approx 3.78$ times, and we measured 3.76 times (0.64 s to 0.17 s).

1.4 Interpretation and recommendations

Each entry is stored with double precision (8 bytes). Full storage $= n_{eqn}^2 \cdot 8 = 10302^2 \cdot 8 \approx 850MB$, while banded storage $= n_{eqn} \cdot b_w \cdot 8 \approx 9MB$. So for larger meshes it is more practical using banded storage. The full solve time increased from about 66 s to 1374 s, while the banded solve time stayed below one second (Figure 1.2.1). Bandwidth minimization roughly halved the bandwidth for meshes numbered along the long side, giving the reduction in solve time discussed earlier. For 60×50 and 50×60, `bandfem` increased the bandwidth to 166, as the original numbering was already near-minimal. A limitation is that there are still stored zeros in the band, and the efficiency depends on the node numbering.

2 Exercise 2 - Virtual Work Principle

2.1 Stress and strain state of the material

The material is only deformed in the x_3-direction, so $\{\boldsymbol{\varepsilon}\} = \begin{bmatrix}0 & 0 & \varepsilon_{33} & 0 & 0 & 0\end{bmatrix}^T$ and $\{\boldsymbol{\sigma}\} = \begin{bmatrix}\frac{\nu}{1-\nu}\sigma_{33} & \frac{\nu}{1-\nu}\sigma_{33} & \sigma_{33} & 0 & 0 & 0\end{bmatrix}^T$. Because $u_1 = u_2 = 0$ and $u_3 = u_3(x_3)$, all fields depend only on x_3 and are uniform over the cross-section, therefore a Cartesian description suffices.

2.2 Strain energy density

The strain energy density is given by $U = \frac{1}{2}\sigma_{ij}\varepsilon_{ij}$. The only non-zero element in the strain vector is ε_{33}. The summation is therefore reduced to $U = \frac{1}{2} \cdot \sigma_{33} \cdot \varepsilon_{33}$. The stress, σ_{33}, is given by $\sigma_{33} = (1-\nu) \cdot c \cdot \varepsilon_{33} = E(x_3) \cdot \varepsilon_{33} \cdot \frac{(1-\nu)}{(1+\nu)(1-2\nu)}$, where $c = \frac{E(x_3)}{(1+\nu)(1-2\nu)}$. This yields equation (1).

$$U = \frac{1}{2} \cdot E(x_3) \cdot \varepsilon_{33}^2 \cdot \frac{(1-\nu)}{(1+\nu)(1-2\nu)} \tag{1}$$

2.3 Conversion to a 1D variational problem

By utilizing that $u_1 = u_2 = 0$ and $u_3 = u_3(x_3)$ we can reduce the expression for the virtual work principle to a 1D problem: $\int_V \delta\varepsilon_{ij}\sigma_{ij}dV = \int_V \delta\varepsilon_{33}\frac{(1-\nu)\cdot E(x_3)}{(1+\nu)(1-2\nu)}\frac{\partial u_3(x_3)}{\partial x_3}dV = \pi R^2 \frac{(1-\nu)}{(1+\nu)(1-2\nu)} \cdot \int_0^H \frac{\partial \delta u_3(x_3)}{\partial x_3} \cdot \frac{\partial u_3(x_3)}{\partial x_3} \cdot E(x_3)dx_3$. Further we can use the fact that the fields are uniform over the cross-section, and rewrite the volume-integral: $\int_V \delta u_i \Phi_i dV = \pi R^2 \rho g \int_0^H \delta u_3(x_3)dx_3$. The surface-integral can be rewritten by utilizing the fact that $\delta u_3 = \delta u_3(0)$ and F_3 are constant over the top of the surface and that $F_3 = \frac{P}{A}$. This gives $\int_{S_F} \delta u_3 F_3 dS = \frac{P}{A} \cdot \delta u_3(0) \cdot \int_{S_F} dS = P \cdot \delta u_3(0)$. This yields:

$$\pi R^2 \frac{(1-\nu)}{(1+\nu)(1-2\nu)} \int_0^H \frac{\partial \delta u_3(x_3)}{\partial x_3}\frac{\partial u_3(x_3)}{\partial x_3}E(x_3)dx_3 = P\delta u_3(0) + \pi R^2 \rho g \int_0^H \delta u_3(x_3)dx_3 \tag{2}$$

The essential boundary conditions are $u_1 = u_2 = 0$ and $u_3(H) = 0$ on S_u which gives $\delta u_1 = \delta u_2 = 0$ and $\delta u_3(H) = 0$ on S_u. The natural boundary conditions are $\sigma_{ij}n_i = F_j$ on S_F, which gives $\sigma_{33} = -p$ and $\sigma_{i3}n_i = 0$ on the die walls.

2.4 Discrete finite element equation

We introduce a general interpolation $u_3(x_3) \approx [\boldsymbol{N}(x_3)]\{\boldsymbol{d}\}$ and $\frac{\partial u_3(x_3)}{\partial x_3} \approx [\boldsymbol{B}(x_3)]\{\boldsymbol{d}\}$. This is used to rewrite the left side of equation (2): $\pi R^2 \frac{(1-\nu)}{(1+\nu)(1-2\nu)} \{\delta \boldsymbol{d}\}^T \int_0^H [\boldsymbol{B}(x_3)]^T [\boldsymbol{B}(x_3)] E(x_3)\, dx_3 \{\boldsymbol{d}\}$. Further the first integral on the right side of equation (2) can be expressed as $P\{\delta \boldsymbol{d}\}^T [\boldsymbol{N}(0)]^T$. The last expression can be rewritten as $\pi R^2 \rho g \{\delta \boldsymbol{d}\}^T \int_0^H [\boldsymbol{N}(x_3)]^T\, dx_3$. We can now cancel the common $\{\delta \boldsymbol{d}\}^T$-term from the total expression and achieve the discrete finite element equation $[\boldsymbol{k}]\{\boldsymbol{d}\} = \{\boldsymbol{f}\}$ shown in equation (3).

$$\pi R^2 \frac{(1-\nu)}{(1+\nu)(1-2\nu)} \int_0^H [\boldsymbol{B}(x_3)]^T [\boldsymbol{B}(x_3)] E(x_3)\, dx_3 \{\boldsymbol{d}\} = P[\boldsymbol{N}(0)]^T + \pi R^2 \rho g \int_0^H [\boldsymbol{N}(x_3)]^T\, dx_3 \tag{3}$$

2.5 Element stiffness matrix for a two-node element

The element stiffness matrix is the term multiplied with $\{\boldsymbol{d}\}$ on the left side of equation (3). For an element between $x_3^{(1)}$ and $x_3^{(2)} = x_3^{(1)} + h_e$ we define $[\boldsymbol{N}(x_3)] = \begin{bmatrix} \frac{x_3^{(2)} - x_3}{h_e} & \frac{x_3 - x_3^{(1)}}{h_e} \end{bmatrix}$, thus $[\boldsymbol{B}] = \frac{1}{h_e} \begin{bmatrix} -1 & 1 \end{bmatrix}$. We then do the matrix multiplication $[\boldsymbol{B}]^T[\boldsymbol{B}]$ in the integral on the left side, set $E(x_3) = E_0 e^{\frac{\beta x_3}{H}}$ and evaluate the integral from $x_3^{(1)}$ to $x_3^{(2)}$, which gives the element stiffness matrix shown in equation (4).

$$[\boldsymbol{k}^e] = \pi R^2 \frac{(1-\nu)}{(1+\nu)(1-2\nu)} \frac{E_0 H}{\beta h_e^2} e^{\beta x_3^{(1)}/H} \left(e^{\beta h_e/H} - 1\right) \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \tag{4}$$

3 Exercise 3 - Isoparametric elements

3.1 Von Mises stress field

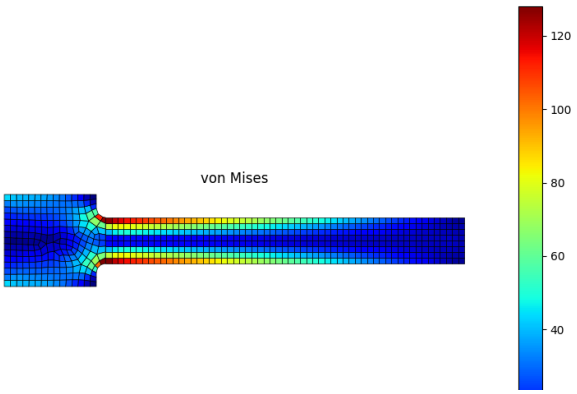


Figure 3.1.1: Von Mises stress field

The lever arm is the distance from the load, and it increases towards the clamped end. Therefore, the bending moment increases the further away from the load it is. Over the height, the stress is highest at the top and bottom edges and zero at the centre. Close to the left end, the bar has twice the height, so the stress there is low. The maximum von Mises stress is 128.01 MPa and is located in the top and bottom fillets. Here the height is halved while the bending moment is still large, and the sudden change in geometry gives a significant stress concentration. Point A corresponds to node 562. The node is shared by two elements, and the von Mises stress at point A is $\sigma_{vM,A} = 118.7\,\text{MPa} < f_y = 250\,\text{MPa}$. The bar does not yield.

3.2 Principal stresses

The shear stress is close to zero along the straight edges, but largest around the fillets. Here the edge is inclined relative to the x- and y-axes, so the stress along the edge also gives a shear component in the x-y system. In the arm, the principal stresses are oriented along the axis of the bar, with tension at the top and compression at the bottom.

These are of equal magnitude, because the bar is only exposed to bending, without any axial force, and the cross-section is symmetrical about the centreline. Along the edges, the principal stresses are parallel to the edge, and along the fillets they follow the curvature. The principal stresses are largest at the fillets, and decrease towards the neutral axis and towards the free end.

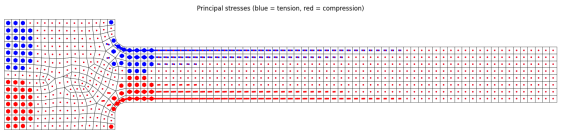


Figure 3.2.1: Principal stresses and directions (Blue = Tension, Red = Compression)

$\sigma_{Bending} = \frac{M_A}{W} = \frac{900\cdot(300-64.45)}{(\frac{1}{12}10\cdot30^3)\cdot\frac{1}{15}} = 141.33$ MPa. The stress concentration factor is $K_t \approx 1.5$ (Pilkey 1997, Chart 3.7, p. 159). The estimated Von Mises for A is then $\sigma_{VM,A} = K_t \cdot 141.33 \approx 212$ MPa.

3.3 Stress in A and tip deflection

Coarse meshes with only one or two elements across the height have relatively low stress in A and a small deflection. The finer the mesh the more deflection and stresses in point A. This occurs because the bending stress is zero on the neutral axis where the stress is evaluated at the element centre. The deflections is small because the bilinear elements experience shear locking in bending making the coarses stiff.

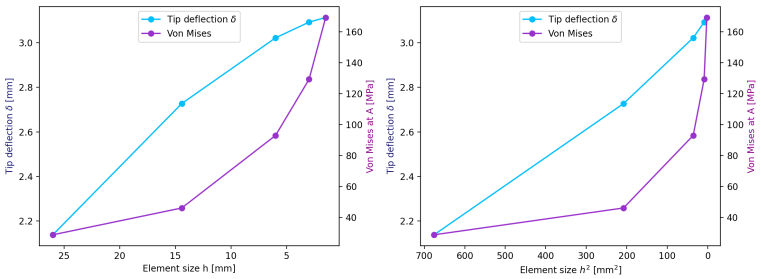


Figure 3.3.1: Tip deflection δ and von Mises stress at A as functions of h and h^2.

3.4 Discussion of differences in stress and deflection

The tip deflection converges faster than the stress at point A, because the stress depends on the displacement derivatives. Looking at the bar as a stepped cantilever beam, we use super position to estimate the tip deflection. $\delta_{tip} = \frac{900}{3\cdot70000}(\frac{300^3-(300-60)^3}{\frac{1}{12}10\cdot60^3} + \frac{(300-60)^3}{\frac{1}{12}10\cdot30^3}) \approx 2.947$ mm. By extrapolating linearly the two finest mesh points, $(h_3 = 2.98, \sigma_{3,A} = 129.19)$ and $(h_{1.5} = 1.49, \sigma_{1.5,A} = 169.13)$, we get $\sigma_{VM,A}(h \to 0) \approx 209$ MPa. This is uncertain because the stresses at A vary significantly between the finest meshes. For $h \to 0$ the FE tip deflection is $\delta \approx 3.12$ mm, about 6 % larger than beam theory.

4 Exercise 4 - Eigenvalue analysis

4.1 Eigenfrequencies

The first six eigenfrequencies are plotted for $x \in [0, 1]$ and shown in figure 4.1.1. For $x = 0$ the column has zero thickness and for $x = 1$ the beam has zero thickness. Zero thickness makes both the stiffness and mass matrices singular. Zero thickness is therefore replaced by $t = 0.001$. We observe that no single x-value minimizes or maximizes all eigenfrequencies. The x-value can therefore only be used to make sure the load frequencies do not match the dominating eigenfrequencies.

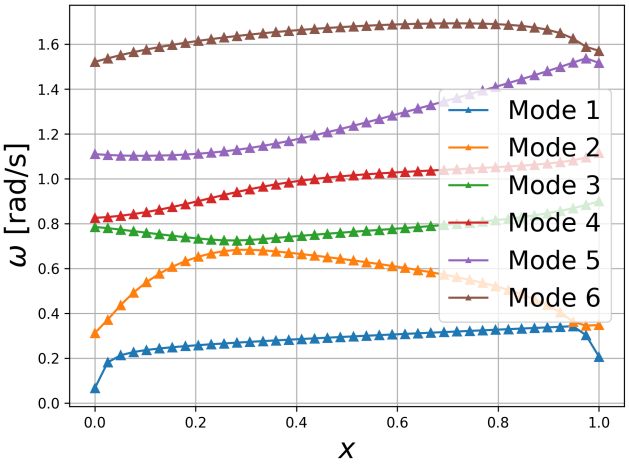


Figure 4.1.1: Eigenfrequencies for mode 1-6 as a function of x.

4.2 Eigenmodes

In figure 4.2.1 eigenmode 1, 2 and 3 are picked out, as the first modes are often the dominating ones. The x-values are chosen as they represent the start- and end values (but not at the singularity), the maximum eigenfrequency for mode 2 and the minimum eigenfrequency for mode 3. The eigenfrequency for mode 1 increases gradually in figure 4.1.1. A thick beam means small deformations in the beam and bigger deformations in the column, and vice versa. The column is shorter than the beam, which gives it more stiffness relative to its mass. This creates the incline for mode 1 in 4.1.1. For mode 2 we observe a maximum for $x \approx 0.3$, which is where the ratio between the modal stiffness and modal mass is largest. In figure 4.2.1 we see that $x = 0,3$ deforms both the column and the beam, and signals a more evenly distributed thickness, which reflects the maximum found in 4.1.1. Mode 3 behaves similary to mode 2, just opposite. It has a minimum approximately where mode 2 has a maximum, as shown in 4.1.1. The thickness distribution that maximizes mode 2 minimizes mode 3. Figure 4.2.1 shows that the beam deforms similarly for $x = 0.3$ and $x = 0.95$, which shows that the thickness distribution for $x = 0.3$ gives little stiffness in relation to its mass, as its thickness is 14 times bigger for $x = 0.3$ than $x = 0.95$.

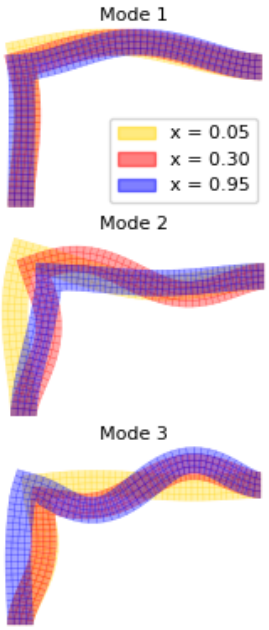


Figure 4.2.1: Mode 1-3 for x = [0.05, 0.3, 0.95]

4.3 Comparison to analytical solutions

We assume that for x close to 0 and 1, the system can be seen as a cantilever beam and a cantilever column, respectively, and we use the analytical solutions for these problems to evaluate the models precision.

x	Mode type	FE	Analytical	Dev. [%]
0.001	Beam, transverse	0.0658	0.051	+30
0.001	Beam, longitudinal	0.7853	0.785	−0.01
0.999	Column, transverse	0.2067	0.203	+1.8
0.999	Column, longitudinal	1.5705	1.571	−0.02

Table 3: FE and analytical eigenfrequencies.

Longitudinal modes perform far better than transversal modes. The longitudinal modes are easier to linearly interpolate, which gives more precise answers. Shear locking for the transversal bending likely overestimates the element stiffness. This is consistent with higher eigenfrequencies for the FE results than the analytical results, which table 3 shows.